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RESEARCH ARTICLE

Spatial patterns of hybridization between bull trout, *Salvelinus confluentus*, and brook trout, *Salvelinus fontinalis* in an Oregon stream network

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Abstract Hybridization with introduced species represents a serious threat to the persistence of many native fish populations. Brook trout (*Salvelinus fontinalis*) have been introduced extensively throughout the native range of bull trout (*S. confluentus*) and hybridization has been documented in several systems where they co-exist and is seen as a significant threat to the persistence of bull trout populations. We identified a group of diagnostic microsatellite loci to differentiate bull trout and brook trout and then used these loci to examine the spatial distribution of hybrids in the Malheur River basin, Oregon USA. In random samples of approximately 100 fish from each of three creeks we identified 181 brook trout, 112 bull trout and 14 hybrids. Although bull trout, brook trout and hybrids were found in all three creeks, they were not evenly distributed; brook trout were primarily found in the lower sections of the creeks, bull trout further upstream, and hybrids in the areas of the greatest overlap. One creek with a population of brook trout in a headwater lake provided an exception to this pattern; brook trout were found distributed throughout the creek downstream of the lake. Several post-F1 hybrids were identified suggesting that hybrids are reproducing in the Malher River Basin. Mitochondrial DNA analysis indicated that both female bull trout and brook trout are involved in hybridization events. Analysis of population structure suggested that brook trout have established multiple spawning populations within the Malheur system. Data presented in this study suggest that relative abundance of brook trout and habitat quality are important factors to consider when evaluating the threat of hybridization to bull trout populations.

Keywords Hybridization · Bull trout · Brook trout · Introgression · *Salvelinus*

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Introduction

Hybridization with nonnative species poses a serious risk of extinction for many native species (Rhymer and Simberloff 1996). Among fish species, hybridization between native and nonnative species has been documented within several families (Scribner et al. 2001). Hybridization among fishes raises several concerns, especially in instances where one (or both) of the species involved is a species of conservation concern. For example, the formation of hybrid swarms between westslope cutthroat trout (*Oncorhynchus clarkii lewisi)* and introduced rainbow trout (*O. mykiss*) and Yellowstone cutthroat trout (*O. c. bouvieri)* has been responsible for the loss of several westslope cutthroat trout

populations (Leary et al. 1995). The threat of hybridization between native and nonnative salmon and trout is of particular concern; many species have been widely introduced outside of their native range and are now in contact with historically isolated but closely related native species (Allendorf et al. 2001).

Bull trout (*Salvelinus confluentus*) and brook trout (*Salvelinus fontinalis*) are two species of salmonids that evolved allopatrically and frequently hybridize when they occur sympatrically. Previous analyses of mitochondrial DNA have demonstrated 4–7% sequence divergence between these two species (Grewe et al. 1990; Taylor et al. 1999). Bull trout are native to northwestern North America, historically ranging from northern California to the Yukon Territory and from the Pacific Coast as far inland as the headwaters of the Saskatchewan River (Cavender 1978; Hass and McPhail 1991). Bull trout have experienced substantial declines in abundance throughout their native range and they are currently listed as threatened in the United States under the US Endangered Species Act (US Fish and Wildlife Service 1999). Brook trout are native to eastern North America, historically ranging from Eastern Manitoba, Ontario, Minnesota and Iowa to the Atlantic Coast (MacCrimmon and Campbell 1969). Since the late 1800s brook trout have been introduced extensively outside their native range. As a result of these introductions brook trout have established naturalized populations throughout the native range of bull trout (MacCrimmon and Campbell 1969). In systems where the two species co-occur, bull trout typically occupy headwater habitats while brook trout typically occupy habitats further downstream with some overlap taking place (Paul and Post 2001; Rieman et al. 2006). When brook trout are introduced into headwater lakes, they are often found more widely dispersed throughout a stream (Adams et al. 2001).

Hybridization between bull trout and brook trout has been documented in several watersheds where the two species co-occur (Leary et al. 1993, 1995; Kitano et al. 1994; Kanda et al. 2002) and is seen as a major threat to the persistence of bull trout populations (Rieman et al. 1997). The degree of hybridization between the two species varies among watersheds (Leary et al. 1995; Kanda et al. 2002). Genetic studies utilizing allozyme markers indicated the majority of bull × brook hybrids appeared to be first generation (F1) and suggested that bull × brook hybrids were mostly sterile (Leary et al. 1993, 1995). More recent evidence from nuclear DNA markers has revealed increased numbers of post-F1 hybrids (F2 and backcrossed fish) although F1 hybrids still appear to be the dominant class (Kanda et al. 2002).

Previous studies of hybridization between bull trout and brook trout focused primarily on genetic species identification of individuals identified in the field as hybrids and did not address the extent of hybridization in specific streams or the spatial distribution of bull trout, brook trout and hybrids in those streams. Recent studies of hybridization between other native and introduced salmonids have demonstrated that oftentimes the pattern of hybridization within a stream network is not random and that the extent of hybridization within a stream can be influenced by a number of biotic and abiotic factors (Weigel et al. 2003; Bettles et al. 2005; Rubidge and Taylor 2005; Boyer et al. 2008; Gunnell et al. 2008). Given these results, a more thorough examination of the spatial patterns of hybridization between bull trout and brook trout is warranted. Information on spatial patterns of hybridization will be important for identifying systems that may be more vulnerable to invasion by brook trout as well as guiding management actions (e.g. placement of barriers, selective removal of brook trout, habitat restoration) for protection or restoration of habitats and local populations of bull trout.

We studied patterns of hybridization between native bull trout and nonnative brook trout in the Malheur River basin in Eastern Oregon, USA. Recent bull trout population surveys in the upper Malheur River documented a number of individuals with intermediate phenotypic characteristics suggesting that hybridization is occurring in this river system. Given the concerns associated with bull trout hybridization, we wished to investigate the degree of hybridization in this system. There were two main objectives of this study. First, we wanted to identify a suite of nuclear microsatellite DNA markers that could be used to distinguish bull trout and brook trout and identify hybrids between the two species. Genetic confirmation of species identification will be useful for verifying the phenotypic characteristics that were used to perform species identification at the time of capture. Secondly, we wanted to use these markers to examine the spatial patterns of hybridization in three tributaries within the Malheur River Basin. Specifically, we were interested in determining the relative abundance of bull trout, brook trout and hybrids and whether the two species constitute separate populations or if there is random mating among the two species. Results of this work will be important for identifying areas where hybridization is most prevalent and establishing management and conservation strategies to benefit bull trout.

Methods

Study location

The Malheur River is a tributary to the Snake River in Eastern Oregon (Fig. 1). There are two populations of bull trout in the Malheur River; one in the headwaters of the North Fork Malheur River and one in the headwaters of the

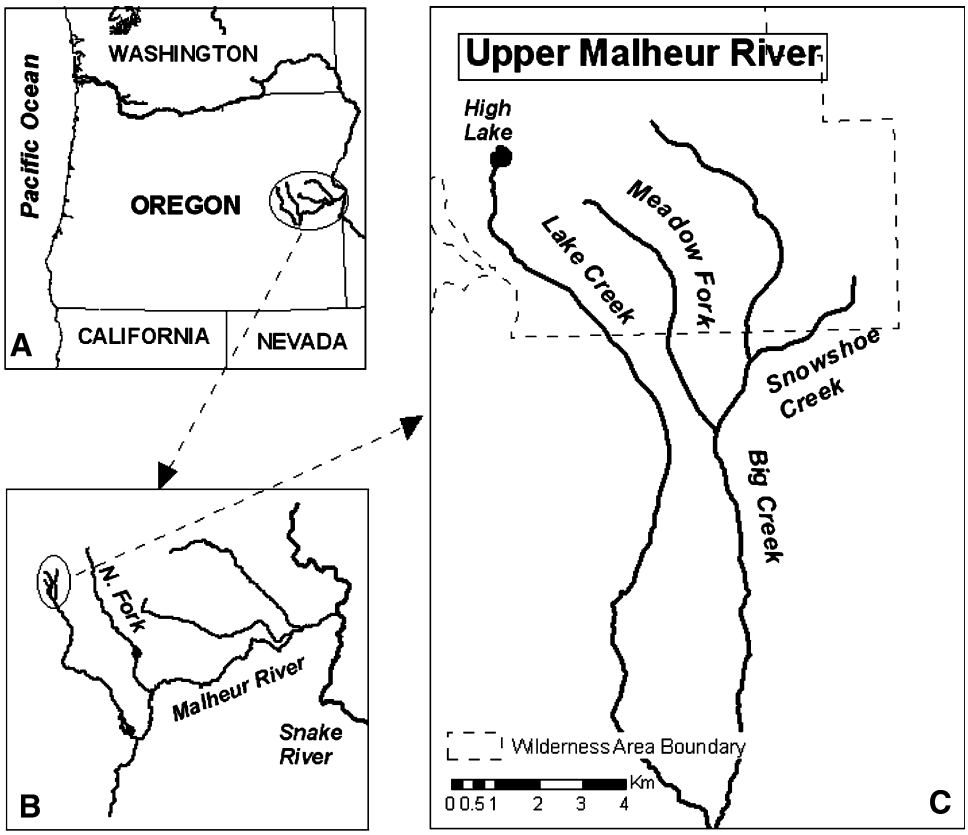

Fig. 1 Study location; the Malheur River Basin, Oregon USA. **A** Shows the location of the Malheur River in Oregon. **B** Shows the location of the upper Malheur River where samples for this study were collected. **C** shows the location of the upper Malheur tributaries that were sampled for this project

upper Malheur River. Both of these populations are located above impassable dams and therefore isolated from one another. Brook trout are thought to have been introduced into the Malheur River system in the 1930s (Schwabe et al. 2004). Presently, naturalized populations of brook trout exist above a waterfall that prevents all upstream movement in Lake Creek, a tributary to the upper Malheur River, and in High Lake, the headwaters for Lake Creek (Fig. 1). These two areas are thought to provide a periodic source of brook trout recruitment for downstream habitat within the upper Malheur River. Recently, fish with intermediate phenotypic characteristics to those of bull and brook trout have been observed in tributaries of the upper Malheur River suggesting that hybridization between native bull trout and introduced brook trout may be occurring (Markle 1992; Schwabe et al. 2004).

Previous population assessments in the upper Malheur River Basin have documented local populations of bull trout in Big Creek, Meadow Fork (a tributary of Big Creek), and Lake Creek (Fig. 1) with both resident and migratory individuals observed (Bowers et al. 1993; Schwabe et al. 2004). Historically bull trout were also observed in Snowshoe Creek but the species was extirpated from this tributary following a wildfire in 1989 and now only brook trout are observed. Radio telemetry has shown that migratory adult bull trout collected downstream of spawning areas in the upper Malheur River primarily utilize the Big Creek system for spawning (Big Creek and Meadow Fork; Schwabe et al. 2001; Fenton et al. 2006). It is suspected that bull trout in Lake Creek are primarily resident fish that are isolated due to current land use practices. Bowers et al. (1993) identified the lower reaches of Lake Creek as a potential thermal barrier to fish passage and irrigation water withdrawals also affect summer rearing bull trout in Lake Creek and disrupt migratory connectivity (Fenton et al. 2006). Recent population surveys found that bull trout outnumbered brook trout 15:1 in Meadow Fork, and a small number of hybrids were believed to be present. In Lake Creek however, brook trout are the dominant species outnumbering bull trout 8:1. Hybrids are believed to be much more prevalent in Lake Creek (Schwabe et al. 2004).

Sample collection

In order to identify a suite of markers that would distinguish bull trout and brook trout, we collected a total of 100 'learning samples': 33 fish that were identified in the field as bull trout, 33 fish that were identified in the field as brook trout and 34 fish that were identified in the field as

putative hybrids. These fish were all collected from Lake Creek. Samples were collected via electrofishing during July 2003 and field biologists identified species using basic phenotypic observations:

Bull trout—bright spotting on a dark back with clear or uniform colored dorsal fin
Brook trout—light green or creamy wavy lines on back with banding present on dorsal fin
Putative hybrid—white spotting present on dorsal fin that extends up ≥50% of the total height of the dorsal fin *or* bright spotting on a dark back with banding present on dorsal fin

We also collected bull trout and brook trout from areas where the two species were allopatric. Thirty bull trout were collected from the upper North Fork Malheur River (Fig. 1) and 30 brook trout were collected from South Pine Creek, in northeastern Iowa, USA.

To examine the spatial patterns of hybridization in the Malheur River Basin we collected approximately 100 fish from each of three creeks: Lake Creek ($n = 108$), Big Creek ($n = 95$) and Meadow Fork ($n = 111$) (Fig. 1). Each creek was divided into an equal number of sites that represented the lower, middle and upper reaches. Sampling start locations were determined by public ownership of land. From the start location, 50 m sites were measured and sampled. One-hundred percent of the wetted channel was sampled upstream of the start location. In this case, sampling efforts did not target specific numbers of pure bull trout, pure brook trout or putative hybrids, but rather represented a random collection of the *Salvelinus* sp. present in each reach of the creeks sampled. A minimum of 30 fish was collected from each reach. In Big Creek, the lower reach was 600 m longer than the middle and upper reaches in order to obtain the minimum requirement of 30 fish per reach. Individuals were collected via electrofishing during July and August 2003 and 2004. Fin clips were taken from all fish collected and preserved in 100% non-denatured ethanol.

Laboratory analyses

We first extracted DNA from the 100 learning samples from Lake Creek, the 30 bull trout from the North Fork Malheur and the 30 brook trout from Pine Creek using a Chelex 100 (Sigma Chemical Co.) resin solution as described by Miller and Kapuscinski (1996). We screened a total of 46 microsatellite markers developed in bull trout, brook trout, rainbow trout (*O. mykiss*) and Dolly Varden (*S. malma*) to identify loci that appeared to distinguish bull trout and brook trout and identify hybrids (Online Appendix). PCR reactions were carried out in 15 µl volumes containing 1X PCR buffer (10 mM Tris–HCl, 50 mM KCl, 1% Triton X-100), 0.2 mM dNTPs, 0.5 µM forward and reverse primer, 1.5 or 2.0 mM $MgCl_2$, and 0.2 U of *Taq* DNA polymerase (Promega Co.). PCR conditions were as follows: an initial DNA denaturation at 94°C for 3 min, followed by 38 cycles of 94°C for 30 s, 30 s at the primer specific annealing temperature, and extension at 72°C for 30 s, and a final annealing and extension at 72°C for 7 min.

Following PCR amplification, reactions were pooled and automated electrophoresis was carried out on an ABI 3130xl Genetic Analyzer (Applied Biosystems Inc., Foster City California) using the GeneScan-500LIZ size standard (Applied Biosystems, Inc.). Electropherograms were analyzed and genotypes were determined for each individual at each locus using the program GENEMAPPER v4.0 (Applied Biosystems, Inc.). We examined the quality of amplification, the distribution of alleles and the allele frequencies for all 46 loci and selected a group of loci that consistently amplified in both species and showed no overlap in allele size between species (i.e. diagnostic markers).

We then extracted DNA from the individuals collected in each of the three creeks to analyze the spatial patterns of hybridization. DNA was extracted from all samples using the Chelex method described above. All samples were then genotyped following the procedures described above using a sub-set of 11 loci we identified to distinguish the two species (see Appendix 1 for PCR conditions). The use of multiple microsatellite loci allowed us to classify hybrid individuals as either F1 hybrids, later generation hybrids (e.g. F2) or as backcrossed individuals with either of the two parental species.

In addition to microsatellite DNA markers, we also amplified a region of the mitochondrial cytochrome B gene for all fish identified as hybrids to determine the direction of hybridization. PCR reactions for mtDNA analysis were conducted in 15 µl volumes and contained 1X PCR buffer (10 mM Tris–HCL, 50 mM KCl, 1% Triton X-100), 0.5 mM dNTPs, 0.5 µM of forward primer 765F (Brown and Thorgaard 2002), 0.5 µM of reverse primer 766R (Brown and Thorgaard 2002), 2.5 mM $MgCl_2$, and 0.2 U of *Taq* DNA polymerase (Promega Co.). PCR conditions were as follows: 38 cycles of 95°C for 45 s, 54°C for 40 s and 70°C for 2 min 30 s, followed by a final extension at 72°C for 3 min. PCR products were then digested overnight with the restriction enzyme *Rsa I* (New England Biolabs) according to the manufactures' instructions. Fragments were separated on 2% agarose gels stained with ethidium bromide and visualized using UV light. Fragment banding patterns produced by each sample were then compared to samples of known bull trout and brook trout from allopatric populations to confirm the diagnostic nature of this marker and to determine the mtDNA lineage for each hybrid fish. Bull trout produced bands of 550, 400,

250 and 50 base pairs and brook trout produced bands of 500, 400, 200, and 150 base pairs (Schriever et al. 2004).

To determine our genotyping error rate, 10% of the samples analyzed plus all individuals genetically identified as hybrids were re-analyzed (i.e. DNA was re-extracted and samples were re-genotyped). A total of 95 samples were re-analyzed for error analysis.

Statistical analyses

We used a simple hybrid index (Campton and Utter 1985; Rubidge and Taylor 2004) to classify individuals as pure bull trout, pure brook trout or bull trout × brook trout hybrids. For each individual the number of brook trout alleles present was summed and divided by the total number of alleles. Individuals possessing all bull trout alleles (hybrid index score of 0.0) were considered pure bull trout and individuals possessing all brook trout alleles (hybrid index score of 1.0) were considered pure brook trout. Individuals with some combination of both bull trout and brook trout alleles were classified as hybrids. Individuals that were heterozygous for a bull trout and a brook trout allele at each locus were considered to be first generation (F1) hybrids (hybrid index score 0.5). Individuals that possessed both bull trout and brook trout alleles and had at least one locus that was homozygous for each parental genotype were considered later generation hybrids (Fn). Individuals that possessed both bull trout and brook trout alleles and had loci that were homozygous for only one of the parental species were considered backcrosses with the parental species (BC_Brook or BC_Bull). These methods were used to classify all of the learning samples as well as the random collections from the three creeks.

For the random collections from each creek, we conducted a chi-squared contingency table analysis to determine if bull trout, brook trout and hybrids were evenly distributed among the three sections of each creek. Because very few individuals of a certain class were observed in some sections of the three tributaries (e.g. 1 hybrid observed in upper Meadow Fork), we performed a Fisher exact test to determine if there were significant differences in species distribution among the three sections of each creek (Zar 1999). We used tests for gametic disequlibrium to determine if the sample of fish from each creek represented multiple spawning populations or if a single *Salvelinus* mating population was present in each creek. Forbes and Allendorf (1991) suggest hybridized populations containing two distinct spawning populations (e.g. bull trout and brook trout) should show evidence of gametic disequilibrium at a high number of loci whereas hybridized populations containing a single mating population (i.e. hybrid swarm) should not show evidence of gametic disequlibrium. Tests for gametic disequilibrium were conducted using the program GENEPOP v4.0 (Raymond and Rousset 1995) and significance values were adjusted for multiple comparisons using a Bonferroni adjustment (Rice 1989).

Although a self sustaining population of brook trout exists in High Lake and Lake Creek, it was unclear if the other two creeks contained additional brook trout spawning populations or if brook trout collected in Big Creek and Meadow Fork were individuals that originated in Lake Creek. We used the Bayesian clustering method of STRUCTURE v2.2 (Pritchard et al. 2000) to investigate the population structure of brook trout in the upper Malheur system. We applied the admixture model that assumes gene flow among populations and allows for correlations of allele frequencies across clusters. This admixture model assigns a proportion of each individual's genome to each of the clusters pursuing solutions that maximize Hardy–Weinberg equilibrium (HWE) and linkage equilibrium within clusters. Using data from all individuals collected during the random sampling efforts that were genetically identified as pure brook trout, 20 replicated unsupervised STRUCTURE runs were performed for each K from 1 to 7. All runs had a burn-in of 30,000 iterations followed by 100,000 iterations. We used the method of Evanno et al. (2005) to infer the correct value of K for the dataset. Evanno et al. (2005) suggested that the method of Pritchard et al. (2000) often leads to an over estimate of K and recommended using the second order rate of change between K and $K + 1$ clusters, Delta K, as a more effective identifier of the correct K for the dataset.

The symmetric similarity coefficient (SSC) was used to determine the similarity of outcomes among the 20 replicate STURUCTURE runs for each K. Using the LargeK-Greedy algorithm of CLUMPP (Jakobsson and Rosenberg 2007) with 1,000 random input sequences we determined the number of distinct modes among the 20 runs at each K by grouping pairs of runs that had a SSC >0.9. Graphical displays of STRUTURE results were generated based on each individual's mean membership in each cluster over the replicate runs.

Following the STRUCTURE analysis, brook trout were grouped into one of three putative clusters (the most likely outcome of the STRUCTURE analysis; see below) based on the cluster they had the greatest level of ancestry from (i.e. maximal q value). Putative clusters were tested for conformance to HWE expectations using an exact test implemented in the program GENEPOP and significance values were adjusted for multiple comparisons using a sequential Bonferroni adjustment (Rice 1989). We estimated measures of genetic variation including mean number of alleles per locus and expected and observed heterozygosity within each putative brook trout population using the program FSTAT v2.9.3.2 (Goudet 2001). To account for differences in

samples size among brook trout populations, we also used FSTAT to estimate allelic richness based on a minimum sample size of 49 individuals. We conducted Wilcoxon's signed rank tests (Zar 1999) to test for significant differences in allelic richness and expected and observed heterozygosity among the putative clusters. FSTAT was also used to estimate the overall level of genetic variation among clusters and the associated 95% confidence level based on 1,000 bootstrap replicates as well as the level of genetic variation among pairs of clusters (F_{ST}; Weir and Cockerham 1984).

The microsatellite markers used in this study were chosen for their ability to distinguish bull trout and brook trout and most of the loci chosen showed little variability within bull trout (Appendix 1). Because of this we did not perform a STRUCTURE analysis to determine the number of bull trout populations present in the upper Malheur River.

Results

Error analysis

A total of 95 samples were re-analyzed for the error analysis. When we excluded missing genotypes due to failed PCR, we found 67 alleles that did not match between the original genotype and the error analysis genotype resulting in an overall error rate of 3.3%. Approximately 33% of these errors were due to allele dropout and 67% were due to genotyping errors. The majority of the genotyping errors occurred in two individuals. These two fish were removed from subsequent analyses. Re-genotyping confirmed the genotypes and classifications of all hybrid individuals.

Selection of loci and analysis of learning samples

Of the 46 microsatellite loci screened, we selected a subset of 11 loci that were easily amplified and scored and exhibited diagnostic differences between bull trout and brook trout (i.e. no overlap in the distribution of alleles). Diagnostic differences were consistent across allopatric and sympatric sampling areas. Our set of loci included: *Omm1128*, *Sco102, Sco104, Sco202, Sco204, Sco215, Sco216, Sco217, Sco218*, *Sfo18* and *SfoC88* (see Appendix 1 for PCR conditions and measures of diversity). Six of the learning samples failed to amplify at more than five loci and were excluded from further analysis. Using these loci we identified 26 pure bull trout and 35 pure brook trout in the group of learning samples. The remaining 33 individuals had a combination of bull trout and brook trout alleles, indicating that hybridization was indeed taking place in Lake Creek. Several of the learning samples had hybrid index scores consistent with backcrossed individuals and later generation hybrids (Fig. 2) and examination of the genotypes of these individuals confirmed the presence of Fn hybrids (hybrid index score 0.71), BC_Brook (hybrid index scores 0.68–0.90) and BC_Bull (hybrid index scores 0.22–0.40).

Of the 33 fish that we identified as hybrids in the group of learning samples, 24 were identified as F1 hybrids, five of which had brook trout mtDNA genotypes and 19 of which had bull trout mtDNA genotypes. Two of the learning samples were identified as a backcross with a bull trout and both of these fish had bull trout mtDNA genotypes. Six of the learning samples were identified as a backcross with a brook trout and all six had brook trout mtDNA genotypes. One of the learning samples was identified as a later generation hybrid and this individual had brook trout mtDNA. Analysis of the allopatric bull trout and brook trout samples confirmed that the mtDNA marker was diagnostic between the two species.

Overall we found that the field identification based on the phenotypic characteristics described above corresponded to the genetic identification 89% of the time. All of the brook trout in the group of learning samples were correctly identified as brook trout in the field. Of the 31 fish we analyzed that were identified in the field as bull trout, 25 were genetically identified as bull trout and 6 of them were genetically identified as hybrids. Four of these six hybrids were identified as F1s and the remaining two were identified as BC_Bull. Of the 31 fish we analyzed that were identified in the field as hybrids, 27 of them were genetically identified as hybrids (F1 and post-F1), four were genetically identified as brook trout and one was genetically identified as a bull trout. In summary, 100% of the brook trout were correctly identified in the field, 80% of the bull trout were correctly identified in the field and 87% of the hybrids were correctly identified in the field.

Spatial distribution of species

Two samples from each creek failed to amplify and were excluded from further analysis. When we examined the species distribution in each of the three creeks, genetic analyses showed that brook trout, bull trout and hybrids were present in all three of the creeks surveyed. In Lake Creek the majority of the fish sampled (89 out of 106) were genetically identified as brook trout and their distribution was nearly equal among the 3 sections of the creek. Bull trout and hybrids were only found only in middle and upper Lake Creek Lake, however (Table 1; Fig. 3). The greatest number of hybrids ($n = 10$) was observed in Lake Creek. The majority of the fish sampled in Big Creek were genetically identified as brook trout (74 of 93). Unlike Lake Creek, the majority of the brook trout in Big Creek

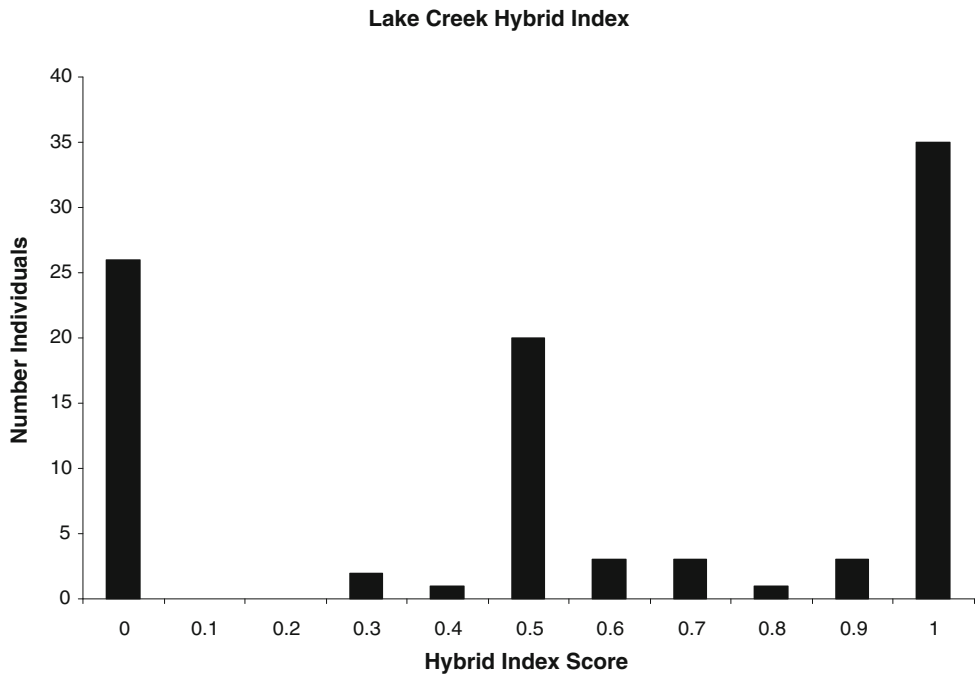

Fig. 2 Hybrid index scores for 94 learning samples collected in Lake Creek. A score of *0.0* indicates that an individual is a pure bull trout and a score of *1.0* indicates an individual is a pure brook trout. The distribution of scores suggests that hybridization does occur and does proceed beyond the F1 generation

were found in the lower and middle sections. There were 18 bull trout in Big Creek, 16 of which were found in the upper section. One fish from upper Big Creek was genetically identified as a hybrid (Table 1; Fig. 3). The majority of the fish sampled in Meadow Fork were genetically identified as bull trout (100 of 109) and were distributed nearly equally among the three sections of the creek (Table 1; Fig. 3). Six fish in Meadow Fork were genetically identified as brook trout and three fish were genetically identified as hybrids (Table 1; Fig. 3).

Of the 14 fish identified as hybrids in our random sample of the three creeks, five were classified as F1 (one from Meadow Fork, four from Lake Creek), six were classified as Fn (two from Meadow Fork, four from Lake Creek) and three were classified as BC_Brook (one from Big Creek, two from Lake Creek; Fig. 3). All of the F1 hybrids had bull trout mtDNA, three of the Fn hybrids had bull trout mtDNA and three had brook trout mtDNA and all of the BC_Brook hybrids had brook trout mtDNA.

Chi-squared contingency table analysis revealed a non-uniform distribution of species classes (bull, brook, hybrid) among the three sections (lower, middle, upper) of all three creeks surveyed (Table 1). In each of the three creeks we observed the following number of locus pairs (out of 55) that showed evidence of gametic disequilibrium: Big Creek 52 pairs of loci, Meadow Fork 53 pairs of loci, and Lake Creek 22 of pairs of loci ($P < 0.001$).

Five clusters (i.e., $K = 5$) had the highest mean probability in the brook trout STRUCTURE analysis (LnP(D) = −5,253.94). However, estimates of ΔK indicated that a K of 3 ($\Delta K = 8.82$) was the appropriate value for the number of brook trout clusters with a second most likely K of 4 ($\Delta K = 4.60$). We plotted proportional membership for individuals assuming a K of 3 (Fig. 4). Results for the 20

Table 1 Chi-squared contingency table analysis of the spatial distribution of bull trout, brook trout and hybrids for each of the three Malheur River tributaries

	Lake Creek*			Big Creek***			Meadow Fork*		
	Bull	Brook	Hybrid	Bull	Brook	Hybrid	Bull	Brook	Hybrid
Lower	0 (2.11)	32 (26.87)	0 (3.02)	1 (6.19)	31 (25.46)	0 (0.34)	25 (28.70)	4 (1.44)	2 (0.86)
Middle	3 (2.58)	32 (32.75)	4 (3.68)	1 (6.19)	31 (25.46)	0 (0.34)	35 (33.33)	1 (1.67)	0 (1.00)
Upper	4 (2.31)	25 (29.39)	6 (3.30)	16 (5.61)	12 (23.08)	1 (0.31)	40 (37.96)	1 (1.90)	1 (1.14)

In each cell the first value represents the observed number of each type and the value in parentheses represents the expected values. Significance levels were evaluated using a Fisher's exact test

* $P < 0.05$; *** $P < 0.0001$

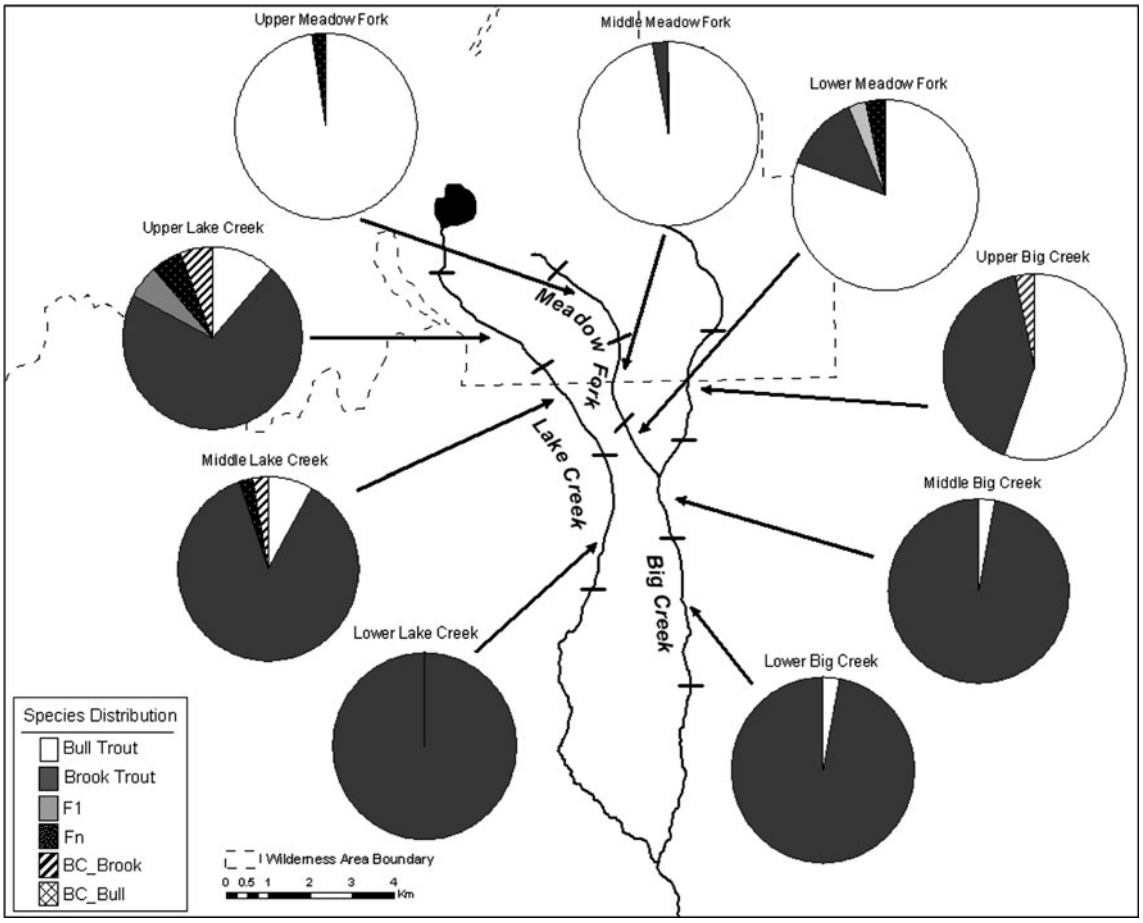

Fig. 3 Spatial distribution of bull trout, brook trout and hybrids in the Malheur River system. *Horizontal lines* through the streams show the approximate boundaries of the different sampling strata (*lower, middle, upper*). The number of fish sampled in each strata can be found in Table 1

replicate STRUCTURE runs with $K = 3$ showed a high degree of consistency (SSC = 0.968). Brook trout with membership to cluster 1 were found almost exclusively in Lake Creek. Individuals with membership to cluster 2 were found in all sampling locations. Fish with membership to cluster 3 were found primarily in Big Creek/Meadow Fork and to a lesser extent in the lower section of Lake Creek (Fig. 4).

When brook trout were grouped into three putative clusters, all three clusters conformed to HWE expectations at all loci. Estimates of variation within the three clusters were similar (Table 2) and we observed no significant differences in any of these measures among the three clusters (Wilcoxon's signed rank tests $P > 0.05$). The overall level of genetic variation among brook trout clusters (i.e. F_{ST}) was 0.055 (95% CI 0.029–0.083). We observed the greatest amount of genetic divergence among clusters 1 and 3 ($F_{ST} = 0.064$; $P < 0.05$) and the least amount of divergence among clusters 2 and 3 ($F_{ST} = 0.046$; $P < 0.05$). The level of divergence among clusters 1 and 2 was 0.055 ($P < 0.05$).

Discussion

Genetic markers have been used to document hybridization between bull trout and brook trout in multiple river systems where the two species co-occur (Leary et al. 1993, 1995; Spruell et al. 2001; Kanda et al. 2002). While some earlier studies were limited by the number and types of markers used (e.g. allozymes, Leary et al. 1993; PINES, Spruell et al. 2001), the use of multiple microsatellite loci in this study allowed us not only to identify individuals with hybrid ancestry, but also to classify individuals as F1, backcross and later generation hybrids, and determine the extent of hybridization in the upper Malheur River. Our analyses demonstrated that hybridization does proceed beyond the F1 generation in the upper Malheur River. Furthermore mtDNA analyses of F1 and Fn hybrids suggest that hybridization events involve both male and female bull trout. Although the two species co-occur in each of the three streams we surveyed, in our random sample from the three streams we documented relatively few hybrids overall (14 hybrids out of 307 fish analyzed) and the species appear to remain genetically distinct.

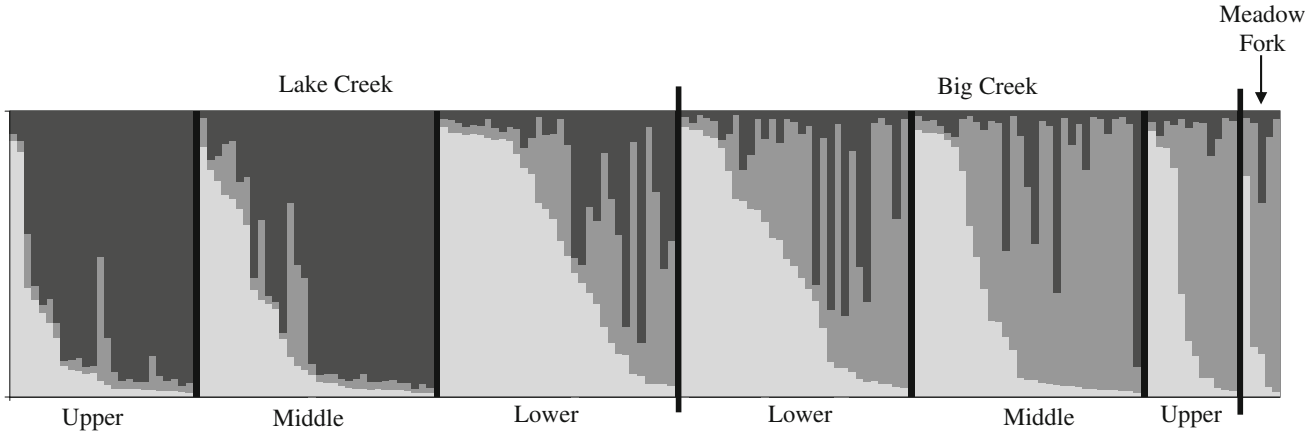

Fig. 4 Output from the program STRUCTURE assuming three ($K = 3$) populations of brook trout in the upper Malheur River system. Each *vertical bar* on the *graph* corresponds to an individual and the different *shaded* regions represent the proportional membership in each of the three clusters. Individuals are grouped by sampling locations and sampling locations are arranged with increasing distance from High Lake (at the headwaters of Lake Creek), the original source of brook trout introduction. Because only five brook trout were observed in Meadow Fork, they were combined into a single group on the *graph*

Table 2 Estimates of genetic variation (based on 11 microsatellite loci) within the three upper Malheur brook trout clusters identified by the program STRUCTURE

Cluster	n	A	A_R	H_{exp}	H_{obs}
1	62	6.909	6.662	0.605	0.612
2	56	7.273	7.194	0.664	0.663
3	50	6.909	6.884	0.656	0.648
Mean		7.030	6.913	0.642	0.641

n Number of individuals; A Mean number of alleles per locus; A_R Allelic richness based on a minimum sample size of 49 individuals; H_{exp} Heterozygosity expected; H_{obs} Heterozygosity observed

Identification and classification of hybrids

Our analyses showed that there was a high correlation between phenotypic (field) and genetic methods of species identification. Among the learning samples from Lake Creek, we found that the genetic identification corresponded to the field identification 89% of the time. This suggests that the criteria established for identifying the different species in the Malheur River Basin are fairly accurate. It is important to consider, however, that this study only tested these identification criteria in the Malheur River Basin and the same criteria may not be as effective when dealing with bull trout and brook trout in other geographic locations. For example, Baumsteiger et al. (2005) found that the percentage of cutthroat trout, steelhead trout and hybrids correctly classified in the field varied substantially among three watersheds where these two species were sympatric.

Hybridization among members of the genus *Salvelinus* is not uncommon (Hammar et al. 1991; Wilson and Hebert 1993; Baxter et al. 1997). While the majority of hybrids documented among *Salvelinus* have been F1s, post-F1 hybrids have been produced for aquaculture (Anderson and Collins 1995) and observed in the wild (Baxter et al. 1997). Previous studies of bull trout × brook trout hybridization documented primarily F1 hybrids between these two species (Leary et al. 1993, 1995). Although post-F1 hybrids have been documented (Kanda et al. 2002), it has been suggested that the majority of the bull trout × brook trout hybrids are sterile (Leary et al. 1995). Between the learning samples and the random samples from each of the three creeks, we identified 47 fish with hybrid ancestry, 18 of which were identified as post-F1. These data indicate that a number of hybrids in the Malheur River system are capable of successful reproduction.

Oftentimes hybrids express phenotypic characteristics that are intermediate to those of the parental species; however, this is not always the case, especially for post-F1 hybrids (Rhymer and Simberloff 1996; Allendorf et al. 2001). Previous studies that examined hybridization between bull trout and brook trout focused on the genetic identification of fish phenotypically identified as hybrids and documented primarily F1 hybrids (Leary et al. 1993, 1995). The fact that our collection efforts targeted not only individuals with intermediate phenotyes but also phenotypic bull trout and brook trout is one possible explanation for the increased number of post F1 hybrids we observed. A number of post-F1 hybrids in this study were mistakenly identified in the field as one of the parental species. Collection efforts that specifically targeted fish with intermediate phenotypes may have missed post-F1 hybrids that phenotypically resembled one of the parental species. Based on our data, it seems that to truly understand the

extent and degree of hybridization in a population, it is best not to limit collections only to individuals that phenotypically appear to be hybrids.

The increased number of post-F1 hybrids we observed in this study may also be related to the sheer abundance of brook trout in the upper Malheur system, specifically in Lake Creek. Recent population surveys found that brook trout outnumber bull trout in Lake Creek by a ratio of 8:1 (Schwabe et al. 2004), presumably due to periodic downstream recruitment from a population of brook trout in a headwater lake. Given the abundance of brook trout and the paucity of bull trout in Lake Creek, spawning opportunities for bull trout would likely favor hybrid matings and backcrosses. It is worth noting that Lake Creek was the only location where we observed individuals classified as BC_Bull and the learning samples also contained a number of BC_Brook. In a study of hybridization in five streams in Montana, Kanda et al. (2002) also observed greater numbers of post-F1 hybrids in a stream where brook trout significantly outnumbered bull trout.

Spatial distribution of species

In our random sample of 307 fish from the three creeks, we found that bull trout, brook trout and hybrids were present in all three creeks; however we observed only 14 hybrid individuals total, the majority of which came from Lake Creek. Although brook trout have been present in the Malheur River system for over 70 years, these data suggest that although the two species co-occur in all three creeks, some isolating mechanism(s) exists that allow the two species to maintain their genetic integrity and distinctiveness. The greatest overlap between the two species was in upper Big Creek where we observed 16 bull trout and 12 brook trout yet only one hybrid. The substantial number of loci that were out of gametic equilibrium in each creek provides further evidence that separate spawning populations exist in these three creeks; i.e. we observed no evidence of a hybrid swarm. These results were similar to those of Leary et al. (1995) and Kanda et al. (2002) who also found that overall the two species appear to maintain their genetic distinctiveness when they co-occur.

Spatial segregation of species in different habitats may facilitate a degree of reproductive isolation between non-native brook trout and native bull trout. In a study of sympatric populations of Dolly Varden and bull trout, Hagen and Taylor (2001) suggested constrained gene flow between these two species was the result of differential habitat use. Studies of spatial segregation between bull trout and brook trout have found when the two species occur sympatrically, brook trout generally occupy the lower regions of the stream and bull trout generally occupy the upper regions (Paul and Post 2001; Rieman et al. 2006). Contingency table analysis showed that the species were not equally distributed in any of the three creeks we surveyed. In Big Creek the dominant species shifted from brook trout in the lower reaches of the stream to bull trout in the upper reaches. In Meadow Fork the majority of the brook trout are also found in the lower reaches of the stream (Table 1; Fig. 3). Bull trout populations are highly sensitive to temperature (Selong et al. 2001; Dunham et al. 2003) as well as habitat modifications (Rieman et al. 1997; Baxter et al. 1999) and laboratory studies have found that brook trout can have a competitive advantage at warmer temperatures (McMahon et al. 2007; Rodtka and Volpe 2007). The upper reaches of all three streams are located at higher elevations within a designated wilderness area and water temperatures and habitat remains relatively pristine, presumably areas that are more favorable for bull trout. Recent surveys of the physical stream conditions in the upper Malheur River basin classified the upper sections of Big Creek and Lake Creek and all of Meadow Fork as nearly 100% of historic conditions. Habitat in the lower reaches of Big Creek was classified as 77% of historic conditions and habitat in lower Lake Creek was classified as 65% of historic conditions (Malheur Watershed Council and Burns Paiute Tribe 2004). Habitat degradation in the lower reaches of the streams we surveyed is another likely cause for the species distribution pattern we observed.

The distribution of species in Lake Creek was somewhat different however; although bull trout were found only in the middle and upper sections (in relatively small numbers), large numbers of brook trout were also found in the upper section of the creek (Table 1; Fig. 3). The difference we observed in the species distribution in Lake Creek is likely due to a self sustaining population of brook trout that exists in the headwaters of the creek (i.e. High Lake; Fig. 1). Adams et al. (2001) found that when brook trout are introduced into a stream via a headwater lake and disperse downstream they tend to be more widely distributed throughout the watershed. Additionally, habitat in the lower reaches of Lake Creek has been modified for irrigation and agricultural purposes (Fenton et al. 2006) and it has been suggested that increased water temperature in the lower section of Lake Creek acts as a thermal barrier for migratory bull trout (Bowers et al. 1993). The combination of these factors may be excluding migratory bull trout from historic spawning grounds in Lake Creek (Schwabe et al. 2004) and preventing bull trout spawning necessary to maintain the population in Lake Creek.

Although High Lake at the headwaters of Lake Creek was the original source of brook trout in the upper Malheur, our results indicate the presence of multiple brook trout populations/clusters within the system. Individuals from one cluster were found primarily in upper and middle Lake

Creek (cluster 1), individuals from a second cluster were found in lower Lake Creek and Big Creek (cluster 2) and individuals in a third cluster were found primarily in Big Creek and Meadow Fork (cluster 3; Fig. 4). Levels of genetic variation among clusters suggested that the least amount of gene flow occurs among clusters 1 and 3 and greater levels of gene flow occur among adjacent clusters. Although we are unaware of physical barriers that would prevent migration of brook trout among upper Malheur River tributaries, thermal barriers and degraded habitat in the lower reaches of Lake Creek that presumably limit bull trout migration may limit brook trout migrations among Lake Creek and Big Creek/Meadow Fork as well. Analysis of brook trout from the original source population in High Lake will be important to fully understand the pattern of brook trout invasion and population structure in the upper Malheur River. Our results illustrate that brook trout in Big Creek and Meadow Fork are not necessarily migrants from upper Lake Creek. Conservation efforts aimed at controlling brook trout in the system should take into account that multiple spawning populations are present and not focus solely on High Lake and upper Lake Creek.

Direction of hybridization

In cases where hybridization is observed between naturally sympatric salmonids, it is often unidirectional (Redenbach and Taylor 2003; Baumsteiger et al. 2005; Kozfkay et al. 2007). Previously Kitano et al. (1994) observed smaller, resident brook trout males "sneaking" fertilizations with larger, migratory bull trout females. Mitochondrial DNA data suggest that this behavioral mechanism is not solely responsible for hybridization in the Malheur River. We observed F1 and Fn hybrids with both bull trout and brook trout mtDNA genotypes illustrating that hybridization proceeds in both directions in the Malheur system. These results were similar to other studies involving hybridization between native and introduced salmonids, including bull trout and brook trout (Kanda et al. 2002; Rubidge and Taylor 2004; Gunnell et al. 2008). Kanda et al. (2002) reported that hybridization between bull trout and brook trout was unidirectional (male brook trout with female bull trout) in three populations, but they observed reciprocal hybridization between bull trout and brook trout in another population that had a higher proportion of brook trout. When two species evolve separately and behavioral isolating mechanisms are weak, demographic factors seem to influence the direction of hybridization (Forbes and Allendorf 1991; Gunnell et al. 2008). The results of our study and others (Kanda et al. 2002) suggest that this is the case for bull trout and brook trout hybridization events, particularly in streams such as Lake Creek where brook trout significantly outnumber bull trout.

Conservation implications

Bull trout are currently listed as a threatened species under the United States endangered species act and hybridization with brook trout is viewed as a major threat to their persistence (US Fish and Wildlife Service 1999). Extinction of local bull trout populations presumably due to hybridization has been documented in systems where large numbers of brook trout were present (Leary et al. 1993; Kanda et al. 2002). In our survey of Lake Creek, brook trout outnumbered bull trout in our random sample by a ratio greater than 10:1 (89 brook trout vs. 7 bull trout). These numbers were consistent with previous population surveys in Lake Creek that found brook trout outnumbered bull trout by a ratio of 8:1 (Schwabe et al. 2004). We also observed greater numbers of hybrids in Lake Creek ($n = 10$) than the other two streams we surveyed. Given that brook trout may display a competitive advantage in some cases when the two species are sympatric (Gunkel et al. 2002; McMahon et al. 2007), interspecific competition has likely contributed to the low abundance of bull trout as well. The likely threats posed by brook trout in Lake Creek suggest that active measures (e.g. construction of barriers to exclude brook trout, selective removal of brook trout; Buktenica 1997) may be needed to ensure long-term persistence of bull trout in this tributary.

In addition to species abundance, there are other important factors that may influence hybridization between bull trout and brook trout. Although brook trout were the predominant species in Big Creek as well, we found only one hybridized individual in Big Creek and three hybrids in Meadow Fork. As mentioned above, habitat in the lower section of Lake Creek has been highly modified and temperatures in sections of Lake Creek are thought to be unsuitable for bull trout. Although habitat in lower Big Creek has also been somewhat degraded, upper Big Creek and Meadow Fork likely provide a refuge for bull trout where habitat remains relatively pristine and, unlike upper Lake Creek, few brook trout are present. The persistence of migratory bull trout has also been recognized as an important factor for the persistence of bull trout populations (Northcote 1997; Rieman and Dunham 2000; Nelson et al. 2002). Although migratory bull trout spawn in Big Creek and Meadow Fork, they are absent from Lake Creek presumably due to seasonal thermal barriers downstream of spawning habitat. In conclusion, the variable pattern of hybridization we observed in the Malheur River can be attributed both to differences in the relative abundance of brook trout and bull trout as well as environmental conditions that may favor one species at the expense of the other. A clear understanding of these different influences can help to guide restoration actions to benefit native bull trout. In the Malheur River, controlling brook trout alone

Table 3 PCR conditions and estimates of diversity for the 11 diagnostic microsatellite loci selected for this study

Locus name	Multiplex set[a]	Flourescent label	T_A	$MgCl_2$ concentration (mM)	Reference/ accession number	*S. confluentus*							*S. fontinalis*						
						Upper North Fork Malheur River		Upper Malheur River		Lake Creek			South Pine Creek		Upper Malheur River		Lake Creek		
						Size range	NA	Size range	NA	H_{exp}	H_{obs}	HWE	Size range	NA	Size range	NA	H_{exp}	H_{obs}	HWE
Omm1128	MS1	VIC	50	1.5	Rexroad et al. (2001)	277–331	5	277–327	4	0.651	0.720	0.269	215–235	3	208–268	10	0.768	0.833	0.536
Sco102	MS1	VIC	55	1.5	AY973576[b]	167–171	2	163–171	3	0.038	0.038	–	129–133	2	129–133	2	0.282	0.333	0.560
Sco204	MS1	NED	60	1.5	DeHaan and Ardren (2005)	185–193	3	189–193	2	0.000	0.000	–	123–139	4	107–185	14	0.873	0.889	0.446
Sco215	MS1	PET	55	1.5	DeHaan and Ardren (2005)	289–293	2	289–293	2	0.498	0.615	0.257	335–345	3	329–349	9	0.493	0.556	0.855
Sfo18	MS1	PET	52	1.5	Angers et al. (1995)	149	1	149	1	0.000	0.000	–	173–181	5	173–187	7	0.720	0.667	0.129
Sco202	MS2	6FAM	50	1.5	DeHaan and Ardren (2005)	127–135	2	127–135	2	0.509	0.654	0.236	104–108	2	104–108	2	0.504	0.528	1.000
SfoC88	MS2	6FAM	58	1.5	AY168192[c]	167	1	167	1	0.000	0.000	–	183–193	4	181–199	7	0.744	0.750	0.406
Sco104	MS2	VIC	55	1.5	AY973578[b]	350–480	8	411–472	7	0.688	0.731	1.000	269–287	4	265–350	15	0.714	0.667	0.388
Sco217	MS2	NED	60	1.5	DeHaan and Ardren (2005)	214–341	7	214–341	8	0.812	0.808	0.220	184	1	168–204	4	0.230	0.250	1.000
Sco216	MS2	PET	55	1.5	DeHaan and Ardren (2005)	241–260	4	241–264	4	0.571	0.692	0.492	135–173	4	131–177	10	0.730	0.750	0.913
Sco218	MS3	PET	55	1.5	DeHaan and Ardren (2005)	222–226	2	222–226	2	0.000	0.000	–	158–174	4	140–218	14	0.873	0.886	0.455

The size range and number of alleles observed at each locus for each species is given separately for allopatric and sympatric populations. Estimates of heterozygosity and conformance to Hardy–Weinberg expectations are based on 26 bull trout and 35 brook trout collected from Lake Creek in 2003

[a] The loci in each multiplex set were amplified separately and combined for electrophoresis following PCR

[b] Ken Warheit and Sewall Young, Washington Department of Fish and Wildlife, personal communication

[c] Tim King, US Geological Survey, personal communication

T_A Locus specific annealing temperature; *NA* number of alleles observed; H_{exp} heterozygosity expected; H_{obs} heterozygosity observed; *HWE* *P*-value for Hardy–Weinberg exact test

may not be sufficient to benefit bull trout. An emphasis on restoring environmental conditions that favor bull trout may be equally important over the long-term.

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Appendix 1

See Table 3.

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